

# What is a constructor?

A constructor is a special block of code that runs automatically when you create an object in Java.



Think of it like the blueprint that sets up your object. Whenever you use the 'new' keyword to create an object, Java calls the constructor.

```
class MyClass {  
    MyClass() {  
        // constructor code  
    }  
}
```

First, the name of the constructor must be exactly the same as the class name.



**Second, it does not have a return type.**  
**Not even void.** This is how Java knows that this is a constructor, not a regular method

# Why do we use constructors?

The main purpose of a constructor is to set initial values to the object's fields. In simple words, it helps us set up the object when it is created.

```
// Parameterized Constructor Example
public class Student {
    String name;
    int age;
    char grade;

    public Student(String n, int a, char g) {
        name = n;
        age = a;
        grade = g;
    }
}

// Creating Student Object
Student s1 = new Student("Ravi", 20, 'A');
```

## Types of constructors in Java

### 1. Default constructor

This is a constructor that takes no arguments. It just assigns some default values to the fields.

```
// Default Constructor Example
public class Dog {
    String breed;

    public Dog() {
        breed = "German Shepherd";
    }
}

// Creating Dog Object
Dog d1 = new Dog();
```

### 2. Parameterized constructor

A parameterized constructor in Java is a constructor that accepts one or more parameters.

```
// Parameterized Constructor Example
public class Student {
    String name;
    int age;
    char grade;

    public Student(String n, int a, char g) {
        name = n;
        age = a;
        grade = g;
    }
}

// Creating Student Object
Student s1 = new Student("Ravi", 20, 'A');
```

# CONSTRUCTOR OVERLOADING

In Java, we can create more than one constructor in the same class, as long as the number or type of parameters is different. This is called **constructor overloading**.

```
// Constructor Overloading
public class Book {
    String title;
    String author;

    public Book() {
        title = "Unknown";
        author = "Unknown";
    }

    public Book(String t) {
        title = t;
        author = "Unknown";
    }

    public Book(String t, String a) {
        title = t;
        author = a;
    }
}
```

## Constructors vs. Methods

| Constructors                                                            | Methods                                                         |
|-------------------------------------------------------------------------|-----------------------------------------------------------------|
| Constructors run automatically when you create an object                | Methods only run when you call them directly.                   |
| Constructors don't have a return type                                   | Methods always have a return type, even if it is void           |
| In simple terms — constructors are used to build and prepare the object | Methods are used to perform actions or behaviors on the object. |

```
public class Student {

    String name;
    int age;

    // Constructor - runs automatically when an object is created
    public Student(String studentName, int studentAge) {
        name = studentName;
        age = studentAge;
        System.out.println("Constructor called: Student created");
    }

    // Method - runs only when called explicitly
    public void displayInfo() {
        System.out.println("Student Name: " + name);
        System.out.println("Student Age: " + age);
    }

    public static void main(String[] args) {
        // Constructor is called automatically here
        Student s1 = new Student("John", 20);

        // Method must be called explicitly
        s1.displayInfo();
    }
}
```